Hi Sir ,
I am Ravikumar,
Naan **Dr.R.Manivannan sir oda student reference-la** indha message panren.
Enoda **Final Year Project Title**:
“IoT Powered Adaptive Sleep Environment for Health & Wellness”.

Description: Sleep quality is a critical component of health and wellness, yet most smart home systems focus on energy efficiency and static appliance control rather than dynamically optimizing sleep conditions. This project introduces an IoT-based adaptive sleep environment that monitors biometric signals such as heart rate, motion, and breathing patterns, and intelligently adjusts temperature, lighting spectrum, airflow, and noise levels in real-time. By learning individual sleep patterns over time, the system creates personalized sleep profiles, predicts disruptions, and proactively optimizes conditions for restorative sleep, transforming a smart home into a health-focused, responsive, and fully integrated sleep ecosystem.

Expected Solution:

- **Real-Time Biometric Monitoring (D):** Sensors track heart rate, motion, and breathing patterns throughout the night. Traditional smart home systems cannot continuously monitor and adjust to these metrics.

- **Personalized Sleep Profiles (D):** The system learns individual sleep patterns over time, creating adaptive profiles that tailor environmental adjustments for each user.

- **Integration with Smart Home (D)**: Communicates with HVAC, smart lights, and bedding systems to enact coordinated adjustments, contrasting with fragmented, manual control of devices.

- **Environmental Optimization (I):** AI adjusts lighting spectrum, temperature, airflow, and noise cancellation dynamically to optimize sleep quality, moving beyond static preset configurations.

- **Predictive Alerts (I):** AI predicts sleep disruptions (e.g., sudden temperature drops, noise spikes) and proactively adjusts conditions or sends recommendations.

*(I) - Independent Outcomes | (D) - Dependent Outcomes

Ippo naan **additional ideas & feature implementation plan** ready pannitu iruken. Ungaloda valuable suggestions venum sir.

Additional Ideas / Features (Proposed):

1. **Sleep Stage Detection** – Heart rate, body movement use panni light / fan / AC automatic-aa adjust aagum.
2. **Environment Control** –
    - Temperature
    - Humidity
    - Light intensity
    - Noise level
      ellam sensor data base-la adaptive-aa control pannum.
3. **Smart Alarm System** – Deep sleep time-la alarm avoid pannitu, light sleep-la gentle-aa alarm trigger aagum.
4. **Mobile App / Web Dashboard** –
    - Real-time sleep data
    - Sleep quality score
    - Daily / weekly report
5. **Health Recommendation Module** – Poor sleep irundha simple tips (sleep timing, room condition) notify pannum.
6. **Cloud Integration** – Data store panni long-term sleep pattern analyze panna mudiyum.

Implementation Plan (Short):

- Sensors: Temperature, Humidity, Pulse, Motion
- Controller: ESP32 / NodeMCU
- Communication: Wi-Fi (IoT)
- Platform: Firebase / ThingSpeak
- Output: Fan, Light, Buzzer control
- App: Android / Web (basic dashboard)